Solution Tutorial V – Payload-Range Diagram

Task 1

Calculate the thrust-specific fuel consumption $TSFC$ of the engines during the cruise and the holding (loiter) flight segment. Use the closest meter value from the standard atmosphere sheet when converting a Flight Level (FL) to meters.

- Cruise:
  Flight Level 330 → $h \approx 10km$
  From standard atmosphere: $\rho = 0.412 \, {}^{kg}/_{m^3}$, $\rho_0 = 1.225 \, {}^{kg}/_{m^3}$

$$TSFC_C = c' \cdot (1 - 0.15 \cdot BPR^{0.65}) \cdot [\, 1 + 0.28 \cdot (1 + 0.063 \cdot BPR^2) \cdot M_\infty \,] \cdot \left(\frac{\rho}{\rho_0}\right)^{0.08}$$

$$\Rightarrow TSFC_C = 0.020 \cdot 10^{-3} \cdot (1 - 0.15 \cdot 9^{0.65}) \cdot [\, 1 + 0.28 \cdot (1 + 0.063 \cdot 9^2) \cdot 0.78 \,] \cdot \left(\frac{0.412}{1.225}\right)^{0.08} = 1.601 \cdot 10^{-5} \, \frac{kg/s}{N}$$

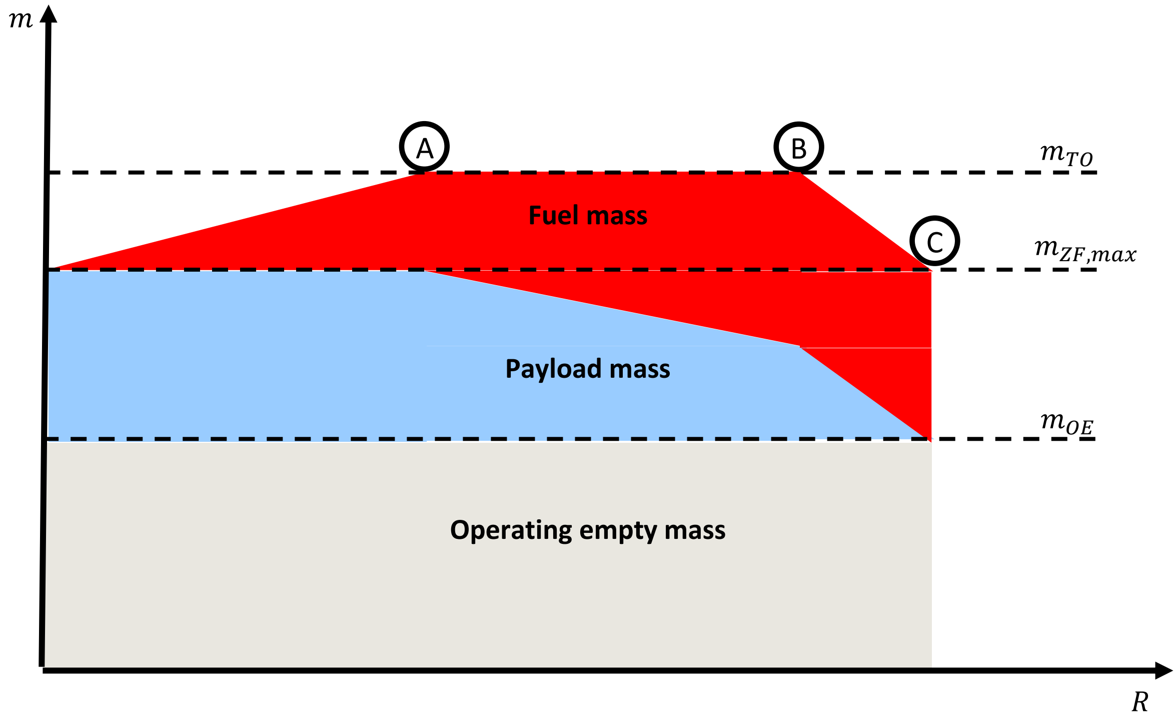
- Loiter:
  Flight Level 100 → $h \approx 3km$
  From standard atmosphere: $\rho = 0.909 \, {}^{kg}/_{m^3}$, $\rho_0 = 1.225 \, {}^{kg}/_{m^3}$

$$TSFC_{lo} = c' \cdot (1 - 0.15 \cdot BPR^{0.65}) \cdot [\, 1 + 0.28 \cdot (1 + 0.063 \cdot BPR^2) \cdot M_\infty \,] \cdot \left(\frac{\rho}{\rho_0}\right)^{0.08}$$

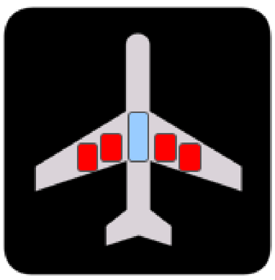
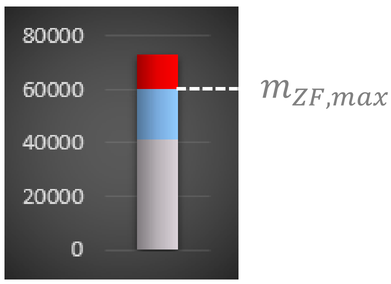
$$\Rightarrow TSFC_{lo} = 0.020 \cdot 10^{-3} \cdot (1 - 0.15 \cdot 9^{0.65}) \cdot [\, 1 + 0.28 \cdot (1 + 0.063 \cdot 9^2) \cdot 0.40 \,] \cdot \left(\frac{0.909}{1.23}\right)^{0.08} = 1.231 \cdot 10^{-5} \, \frac{kg/s}{N}$$

Task 2

Examine the payload-range diagram on the next page. What payload mass in kg can be transported in each one of the points A, B and C? Define the take-off mass m_{TO} and landing mass m_{LND} for each point. (Fuel reserves at landing can be neglected in a first approximation).

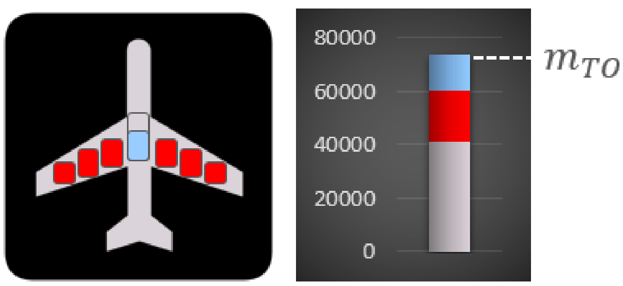


- <u>Point A:</u>

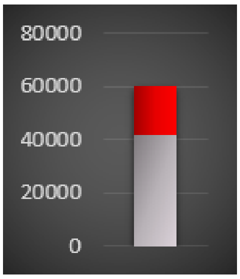
Maximum payload is carried: $m_{PL} = m_{PL,max} = 19200\ kg$
Maximum take-off mass is reached: $m_{TO} = m_{TO,max} = 73500\ kg$
Landing mass is thus: $m_L = m_{OE} + m_{PL,max} = 41300\ kg + 19200\ kg = 60500\ kg$

- <u>Point B:</u>



Payload for full tanks is carried: $m_{PL} = m_{PL@fuel,max} = 13500\ kg$
Maximum take-off mass is achieved: $m_{TO} = m_{TO,max} = 73500\ kg$
Landing mass is thus: $m_L = m_{OE} + m_{PL@fuel,max}$
$= 41300\ kg + 13500\ kg = 54800\ kg$

- Point C:

No payload is carried: $m_{PL} = 0\ kg$
Take-off mass is max. fuel mass plus empty mass: $m_{TO} = m_{OE} + m_{fuel,max}$
$= 41300\ kg + 18700\ kg = 60000\ kg$
Landing mass is thus: $m_L = m_{OE} = 41300\ kg$

Task 3

What is the tanked fuel fraction m_{fuel}/m_{TO} and the actual fuel mass m_{fuel} in each point? Use the relation $m_{fuel} = m_{TO} - m_L$ assuming all fuel is used prior to landing.

- Point A:
Tanked fuel fraction:
$$m_{fuel}/m_{TO} = (m_{TO} - m_L)/m_{TO} = (73500 - 60500)/73500 = 0.1769$$
Tanked fuel mass:
$$m_{fuel} = m_{TO} - m_L = 73500 - 60500 = 13000\ kg$$

- Point B:
Tanked fuel fraction:
$$m_{fuel}/m_{TO} = (m_{TO} - m_L)/m_{TO} = (73500 - 54800)/73500 = 0.2544$$
Tanked fuel mass:
$$m_{fuel} = m_{TO} - m_L = 73500 - 54800 = 18700\ kg$$

- Point C:
Tanked fuel fraction:
$$m_{fuel}/m_{TO} = (m_{TO} - m_L)/m_{TO} = (60000 - 41300)/60000 = 0.3117$$
Tanked fuel mass:
$$m_{fuel} = m_{TO} - m_L = 60000 - 41300 = 18700\ kg$$

Task 4

Calculate the weight fraction during the holding (loiter) segment w_{76}.

Apply the Breguet equation for loiter flight:

$$\Delta t = \frac{1}{\mathrm{g} \cdot \mathrm{TSFC}_{lo}} \cdot \left(\frac{\mathrm{C}_L}{\mathrm{C}_D}\right)_{lo} \cdot ln\frac{w_6}{w_7}$$

$$\Longrightarrow ln\frac{w_6}{w_7} = \frac{\Delta t \cdot \mathrm{g} \cdot \mathrm{TSFC}_{lo}}{\left(\frac{\mathrm{C}_L}{\mathrm{C}_D}\right)_{lo}}$$

$$\Longrightarrow \frac{w_7}{w_6} = w_{76} = \mathrm{e}^{\left[-\frac{\Delta t \cdot \mathrm{g} \cdot \mathrm{TSFC}_{lo}}{\left(\frac{\mathrm{C}_L}{\mathrm{C}_D}\right)_{lo}}\right]}$$

$$\Longrightarrow w_{76} = \mathrm{e}^{\left[-\frac{2700 \cdot 9.81 \cdot 1.231 \cdot 10^{-5}}{10}\right]} = 0.9679$$

Task 5

Calculate the weight fraction during the cruise segment w_{54} in each point by using the empirical mass fractions and the loiter mass fraction in combination with the individual calculated take-off and landing masses.
Remember that: $m_{TO} \cdot (w_{10} \cdot w_{21} \cdot w_{32} \cdot w_{43} \cdot w_{54} \cdot w_{65} \cdot w_{76} \cdot w_{87}) = m_L$

From table “transport jets”:

$$w_{10} = 0.990$$
$$w_{21} = 0.990$$
$$w_{32} = 0.995$$
$$w_{43} = 0.980$$
$$w_{65} = 0.990$$
$$w_{87} = 0.995$$

Rearranging given equation:

$$m_{TO} \cdot (w_{10} \cdot w_{21} \cdot w_{32} \cdot w_{43} \cdot w_{54} \cdot w_{65} \cdot w_{76} \cdot w_{87}) = m_L$$

$$\Longrightarrow w_{54} = \frac{m_L}{m_{TO} \cdot (w_{10} \cdot w_{21} \cdot w_{32} \cdot w_{43} \cdot w_{65} \cdot w_{76} \cdot w_{87})}$$

A. Applying equation for point A:

$$w_{54} = \frac{60500}{73500 \cdot (0.990 \cdot 0.990 \cdot 0.995 \cdot 0.980 \cdot 0.990 \cdot 0.9679 \cdot 0.995)} = 0.9034$$

B. Applying equation for point B:

$$w_{54} = \frac{54800}{73500 \cdot (0.990 \cdot 0.990 \cdot 0.995 \cdot 0.980 \cdot 0.990 \cdot 0.9679 \cdot 0.995)} = 0.8182$$

C. Applying equation for point C:

$$w_{54} = \frac{41300}{60000 \cdot (0.990 \cdot 0.990 \cdot 0.995 \cdot 0.980 \cdot 0.990 \cdot 0.9679 \cdot 0.995)} = 0.7554$$

Task 6

Finally, derive the cruise range from the weight fraction w_{54} for each point A, B, C and draw the payload-range diagram.

Breguet equation for cruise flight:

$$R = \frac{V_C}{g \cdot TSFC_C} \cdot \left(\frac{C_L}{C_D}\right)_C \cdot ln\frac{w_4}{w_5} = - \frac{V_C}{g \cdot TSFC_C} \cdot \left(\frac{C_L}{C_D}\right)_C \cdot ln\frac{w_5}{w_4} \qquad \left(\frac{w_5}{w_4} = w_{54}\right)$$

Cruise: M_C = 0.78, Flight Level 330 → $h \approx 10km$
From standard atmosphere: $a = 299\ {}^{m}/_{s}$
$\longrightarrow V_C = M_C \cdot a = 0.78 \cdot 299 = 233.2\ m/s$

- Applying equation for point A:

$$\mathrm{R} = - \frac{V_C}{\mathrm{g} \cdot \mathrm{TSFC}_C} \cdot \left(\frac{C_L}{C_D}\right)_C \cdot ln\frac{w_5}{w_4} = - \frac{233.2}{9.81 \cdot 1.601 \cdot 10^{-5}} \cdot 16.3 \cdot \ln(0.9034)$$
$$= 2459\ km$$

- Applying equation for point B:

$$\mathrm{R} = - \frac{V_C}{\mathrm{g} \cdot \mathrm{TSFC}_C} \cdot \left(\frac{C_L}{C_D}\right)_C \cdot ln\frac{w_5}{w_4} = - \frac{233.2}{9.81 \cdot 1.601 \cdot 10^{-5}} \cdot 16.3 \cdot \ln(0.8182)$$
$$= 4856\ km$$

- Applying equation for point C:

$$\mathrm{R} = - \frac{V_C}{\mathrm{g} \cdot \mathrm{TSFC}_C} \cdot \left(\frac{C_L}{C_D}\right)_C \cdot ln\frac{w_5}{w_4} = - \frac{233.2}{9.81 \cdot 1.601 \cdot 10^{-5}} \cdot 16.3 \cdot \ln(0.7554)$$
$$= 6789\ km$$

Draw points in payload-range diagram: